

Christian Etherson

COLLEGE FOOTBALL

PREDICTION MODELS



OVERVIEW

Extension of CFB Ranking System Project

- Goal: Build on the logistic regression model from the previous project to predict college football games accurately.
- Goal: Use train/test splits to obtain a more robust measure of accuracy
- Goal: Test out various classification models
- Goal: Predict this weekends championships games and compare with reputable prediction outlets

OVERVIEW

Recap of Last Project

- Dataset From kaggle
- Filtered game data from 2010 to 2025
- Final Logistic Regression Model Included:
 - Season avg. turnover rate difference
 - Season avg. third-down conversion rate difference
 - Season avg. margin of victory difference

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.452735   0.024042  18.831  < 2e-16 ***
## turnovers_pg_diff_season -0.157077  0.042023  -3.738 0.000186 ***
## third_down_rate_diff_season 0.700035  0.354523   1.975 0.048316 *
## mov_diff_season      0.118431  0.002663  44.466  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 16921  on 12741  degrees of freedom
## Residual deviance: 10848  on 12738  degrees of freedom
## (4 observations deleted due to missingness)
## AIC: 10856
```

PREDICTION RESULTS- 71% ACCURATE

	## 1	Kansas St.	Oklahoma St.	0.983		## 21	Memphis	East Carolina	0.651
	## 2	North Texas	UAB	0.981		## 22	Texas	Georgia	0.621
	## 3	Georgia Tech	Boston College	0.980		## 23	Oregon St.	Tulsa	0.598
	## 4	Utah	Baylor	0.960		## 24	TCU	BYU	0.575
	## 5	UTSA	Charlotte	0.952		## 25	Wyoming	Fresno St.	0.531
	## 6	Marshall	Georgia St.	0.944		## 26	Iowa	Southern California	0.520
	## 7	Louisiana	ULM	0.915		## 27	Kentucky	Western Ky.	0.517
	## 8	San Jose St.	Nevada	0.906		## 28	Maryland	Illinois	0.493
	## 9	Arkansas	LSU	0.901		## 29	Oklahoma	Alabama	0.487
	## 10	Michigan	Northwestern	0.880		## 30	Boise St.	San Diego St.	0.467
	## 11	South Fla.	Navy	0.880		## 31	North Carolina	Wake Forest	0.430
	## 12	Penn St.	Michigan St.	0.876		## 32	Mississippi St.	Missouri	0.325
	## 13	Louisiana Tech	Washington St.	0.870		## 33	West Virginia	Arizona St.	0.303
	## 14	Notre Dame	Pittsburgh	0.804		## 34	Texas St.	Southern Miss.	0.300
	## 15	Liberty	FIU	0.786		## 35	Fla. Atlantic	Tulane	0.287
	## 16	Eastern Mich.	Ball St.	0.786		## 36	Air Force	UConn	0.240
	## 17	Virginia	Duke	0.772		## 37	New Mexico St.	Tennessee	0.220
	## 18	Kennesaw St.	Jacksonville St.	0.705		## 38	Colorado St.	New Mexico	0.194
	## 19	Coastal Carolina	Ga. Southern	0.668		## 39	Purdue	Washington	0.154
	## 20	Arizona	Cincinnati	0.658		## 40	NC State	Miami (FL)	0.142
						## 41	App State	James Madison	0.138
						## 42	South Carolina	Texas A&M	0.0978
						## 43	UCF	Texas Tech	0.0627
						## 44	UCLA	Ohio St.	0.0222
						## 45	Wisconsin	Indiana	0.0136

IMPROVEMENTS

Areas of improvement for last project

- Include defensive statistics
- Address Margin of Victory
- Robust testing method

NEW PROJECT

Variables

- Created variables to test:
- Avg. Pass Efficiency Difference
- Avg. Penalty per Game Difference
- Avg. Rush Efficiency Difference
- Avg. First Downs per Game Difference
- Avg. Third Down Conversion Rate Difference
- Avg. Turnovers per Game Difference
- Avg. Margin of Victory Difference
- ★ Avg. Yards Allowed per Game Difference
- ★ Avg. Turnovers Caused per Game Difference
- ★ Avg. Truncated Margin of Victory Difference

MARGIN OF VICTORY

Variables

★ Avg. Truncated Margin of Victory

Difference:

- Mutates individual margin of victory stats to fit within $[-4, 4]$
- Ex: MOV of 25 is reduced to 4,
- MOV of -10 is reduced to -4
- Season average is calculated with those truncated values
- Produced the highest test accuracy values

FIRST MODEL

Non Truncated MOV Full Model

- Split train/test data into 2010-2022 for training and 2023-2024 for testing
- Resulted in roughly 85/15 split
- AIC: 9067.4
- Test Accuracy: 78.84%

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.4456505	0.0263584	16.907	< 2e-16	***
pass_eff_diff_season	0.0124125	0.0214470	0.579	0.56276	
rush_eff_diff_season	-0.0381855	0.0283397	-1.347	0.17784	
turnovers_pg_diff_season	-0.1363687	0.0477409	-2.856	0.00428	**
pen_eff_diff_season	0.0375827	0.0247149	1.521	0.12835	
first_downs_diff_season	-0.0166286	0.0108341	-1.535	0.12482	
third_down_rate_diff_season	1.2702614	0.4766477	2.665	0.00770	**
mov_diff_season	0.1221203	0.0051338	23.787	< 2e-16	***
yards_allowed_diff_season	0.0008160	0.0005814	1.403	0.16048	
turnovers_caused_pg_diff	0.0648543	0.0510353	1.271	0.20381	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

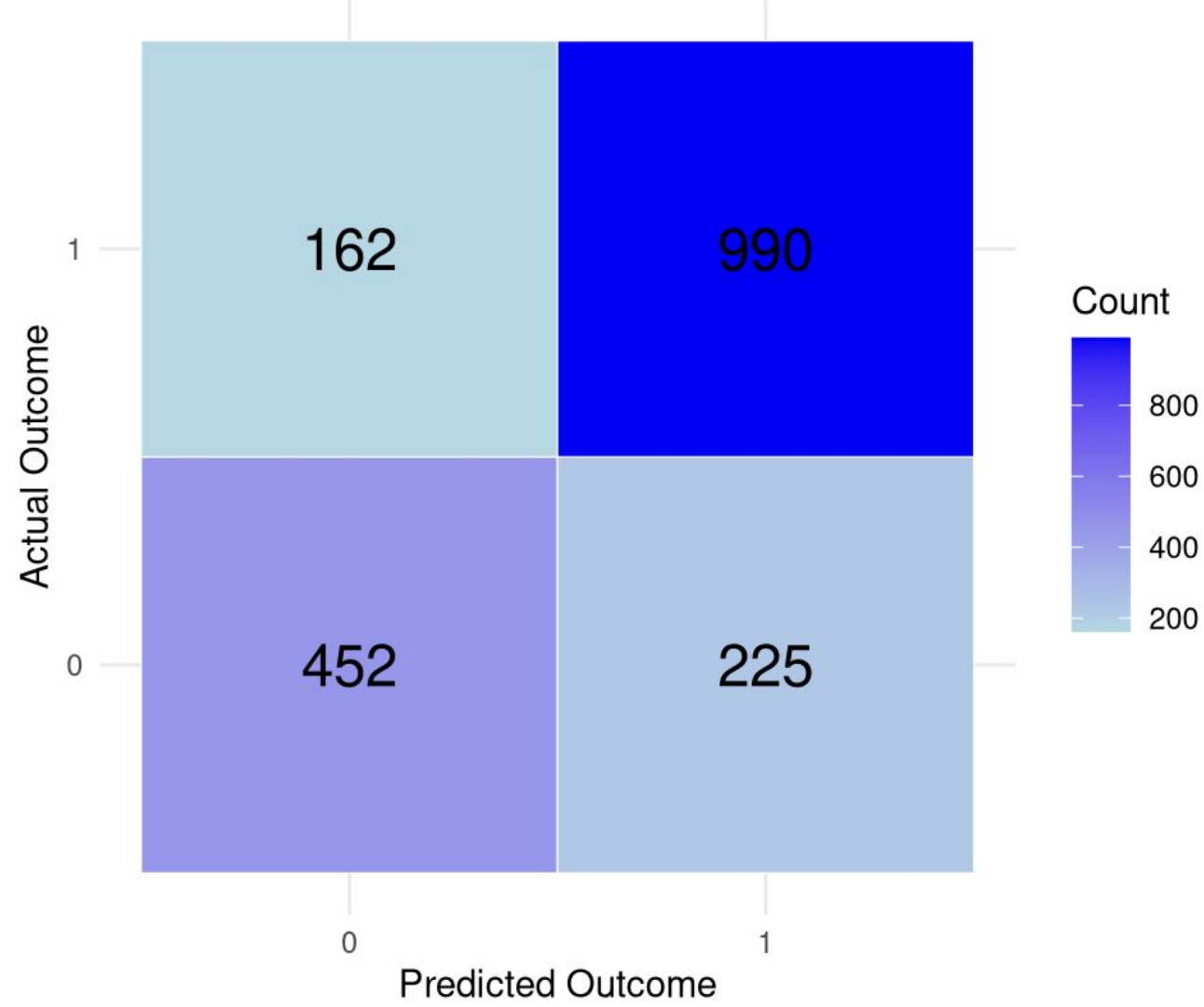
Null deviance: 14236.2 on 10698 degrees of freedom
Residual deviance: 9047.4 on 10689 degrees of freedom
AIC: 9067.4

Number of Fisher Scoring iterations: 6

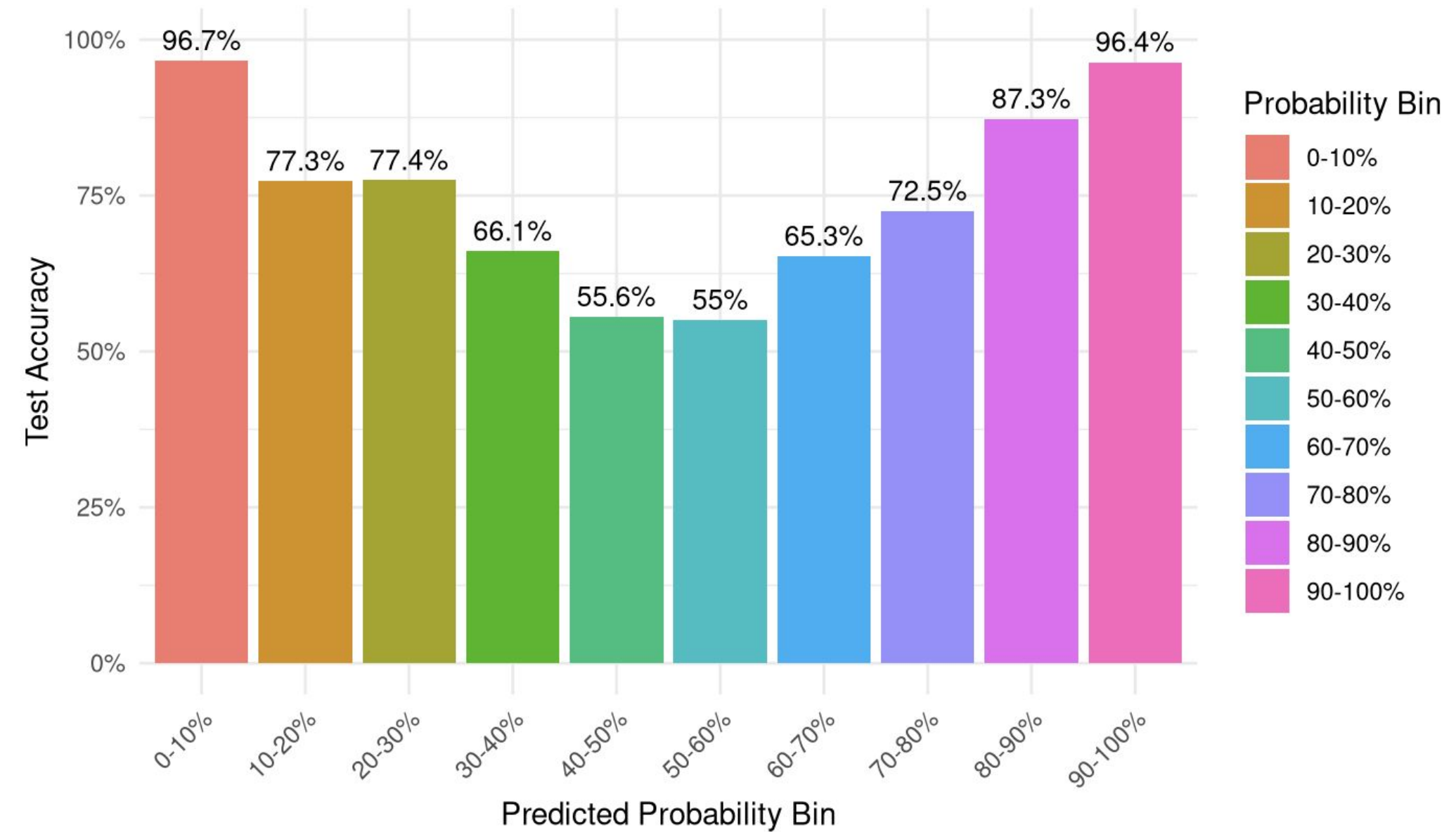
FIRST MODEL

Non Truncated MOV Full Model

Confusion Matrix for Full Non Truncated MOV Model



Binned Test Accuracies for Full Non Truncated MOV Model



- Wins Correctly Predicted: 85.94%, Losses Correctly Predicted: 66.77%

SECOND MODEL

Truncated MOV Full Model

- Used the truncated MOV variable instead of the normal MOV
- AIC: 8101.2 (down from 9067.4)
- Test Accuracy: 81.68% (up from 78.84%)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.4672720	0.0281720	16.586	<2e-16	***
pass_eff_diff_season	-0.0053139	0.0223759	-0.237	0.8123	
rush_eff_diff_season	-0.0063592	0.0290685	-0.219	0.8268	
turnovers_pg_diff_season	-0.0112243	0.0514347	-0.218	0.8273	
pen_eff_diff_season	-0.0037710	0.0272400	-0.138	0.8899	
first_downs_diff_season	0.0052167	0.0105853	0.493	0.6221	
third_down_rate_diff_season	-0.2105313	0.5281348	-0.399	0.6902	
mov_trunc_diff_season	0.9467050	0.0277814	34.077	<2e-16	***
yards_allowed_diff_season	0.0002932	0.0004937	0.594	0.5526	
turnovers_caused_pg_diff	-0.0876201	0.0530425	-1.652	0.0986	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

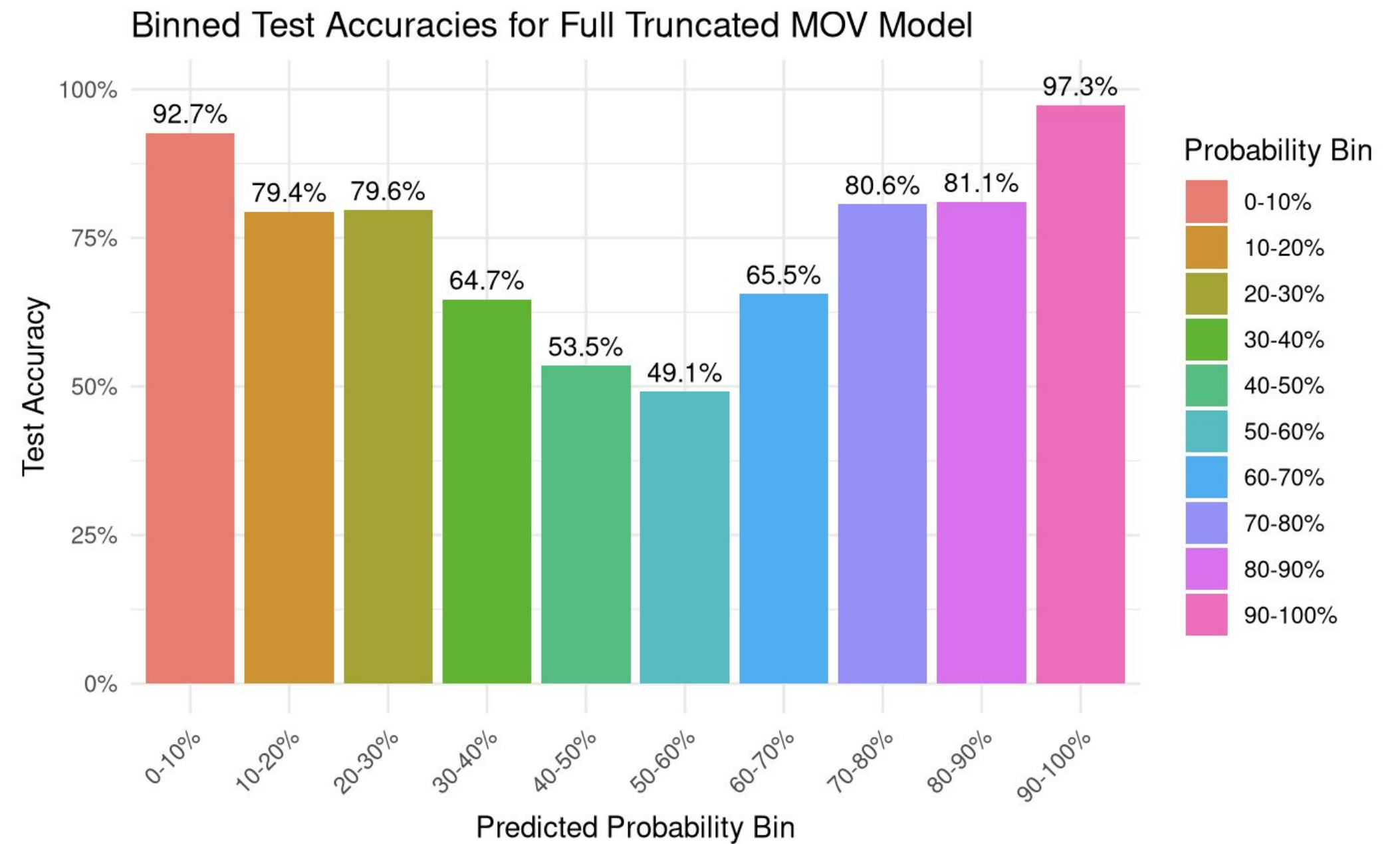
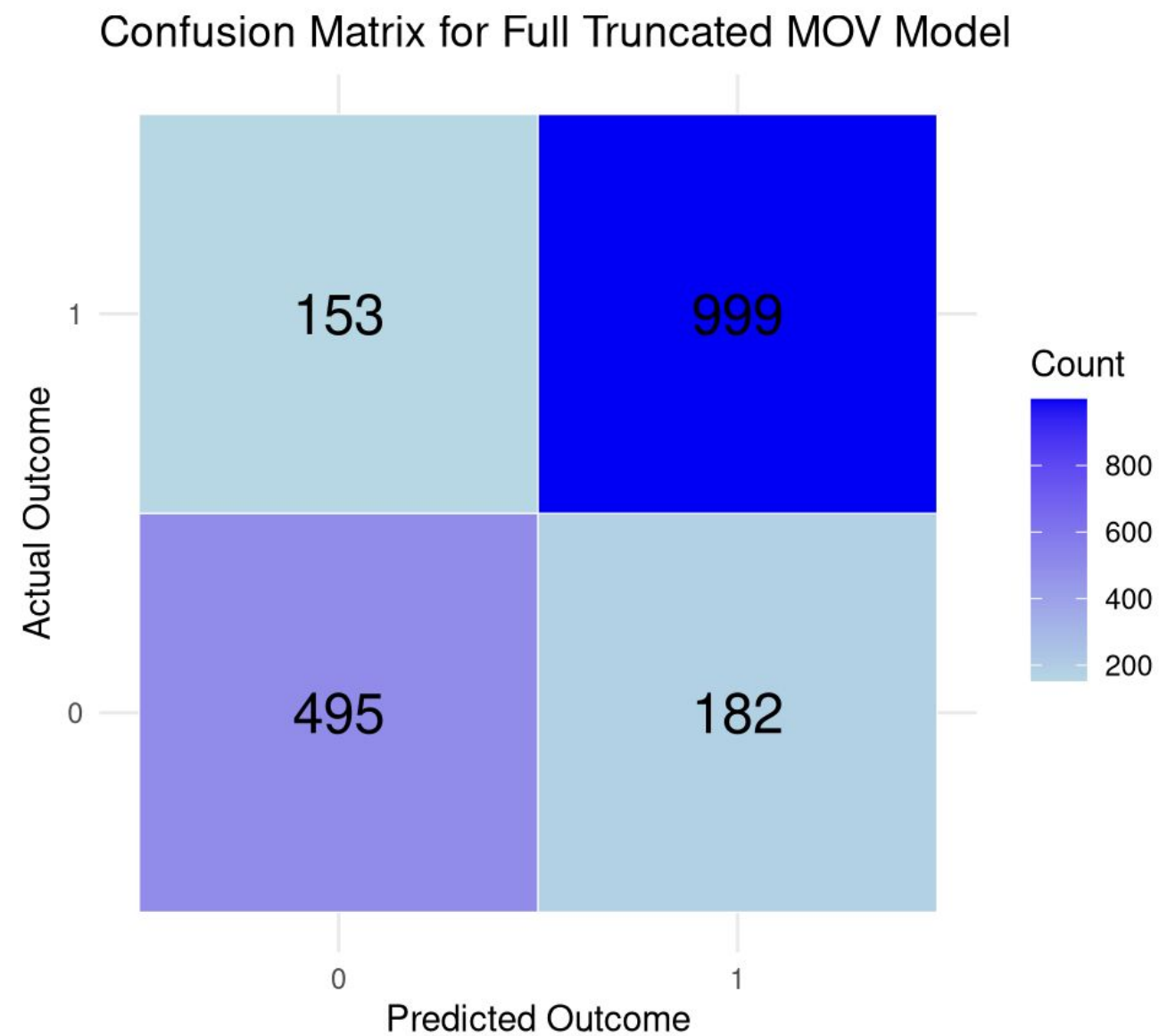
(Dispersion parameter for binomial family taken to be 1)

Null deviance: 14236.2 on 10698 degrees of freedom
Residual deviance: 8081.2 on 10689 degrees of freedom
AIC: 8101.2

Number of Fisher Scoring iterations: 6

SECOND MODEL

Truncated MOV Full Model



- Wins Correctly Predicted: 86.72%, Losses Correctly Predicted: 73.12%

THIRD MODEL

Truncated MOV Reduced Model

- Used the truncated MOV variable and reduced the number of variables until test accuracy gains stalled.
- Kept non-significant variables with the highest z-score
- AIC: 8094 (down from 8101.2)
- Test Accuracy: 81.79% (up from 81.68%)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.4677604	0.0281514	16.616	<2e-16	***
first_downs_diff_season	0.0048588	0.0102605	0.474	0.636	
third_down_rate_diff_season	-0.0725697	0.4950993	-0.147	0.883	
mov_trunc_diff_season	0.9312348	0.0235170	39.598	<2e-16	***
yards_allowed_diff_season	0.0002266	0.0004826	0.469	0.639	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

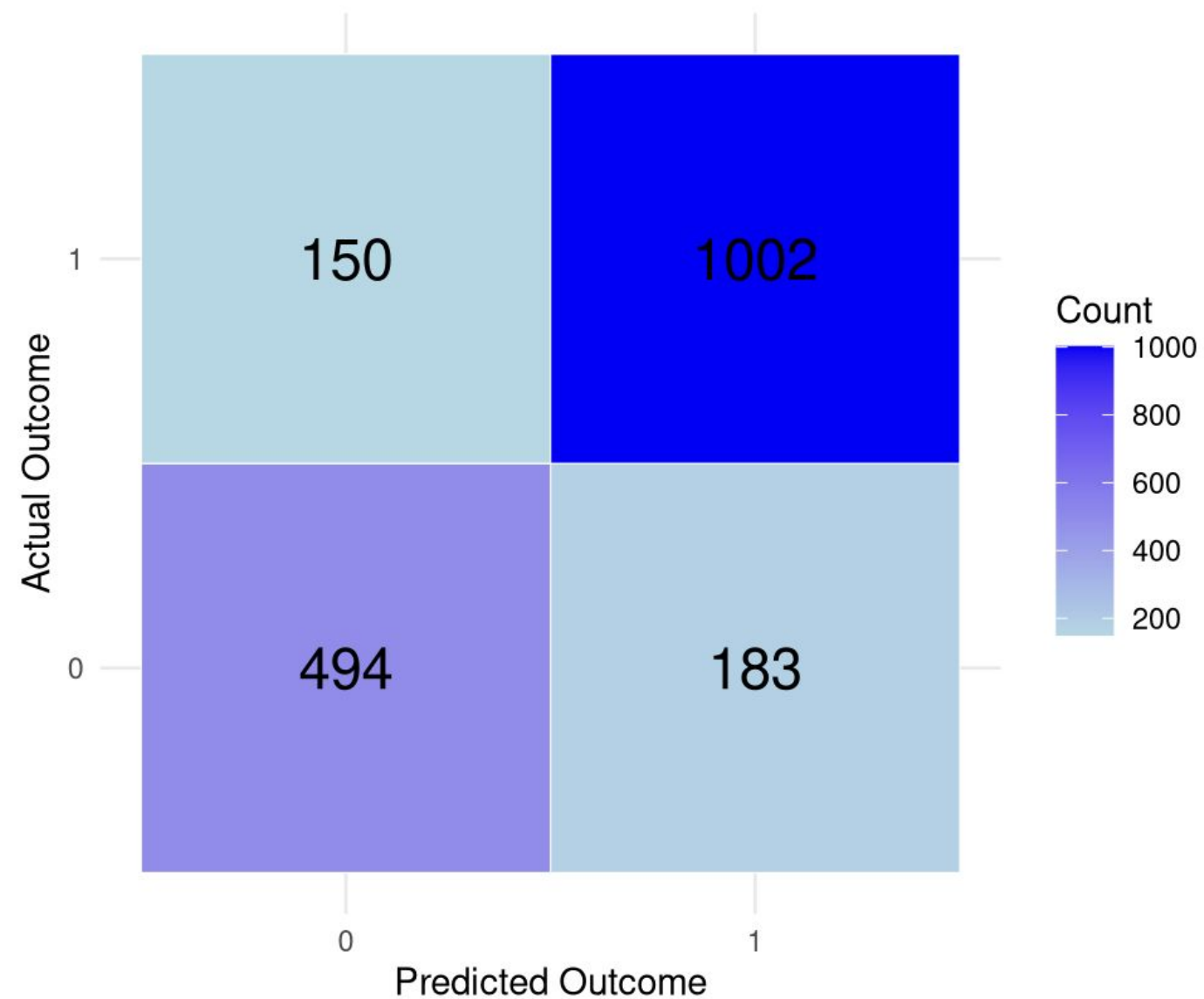
Null deviance: 14236 on 10698 degrees of freedom
Residual deviance: 8084 on 10694 degrees of freedom
AIC: 8094

Number of Fisher Scoring iterations: 6

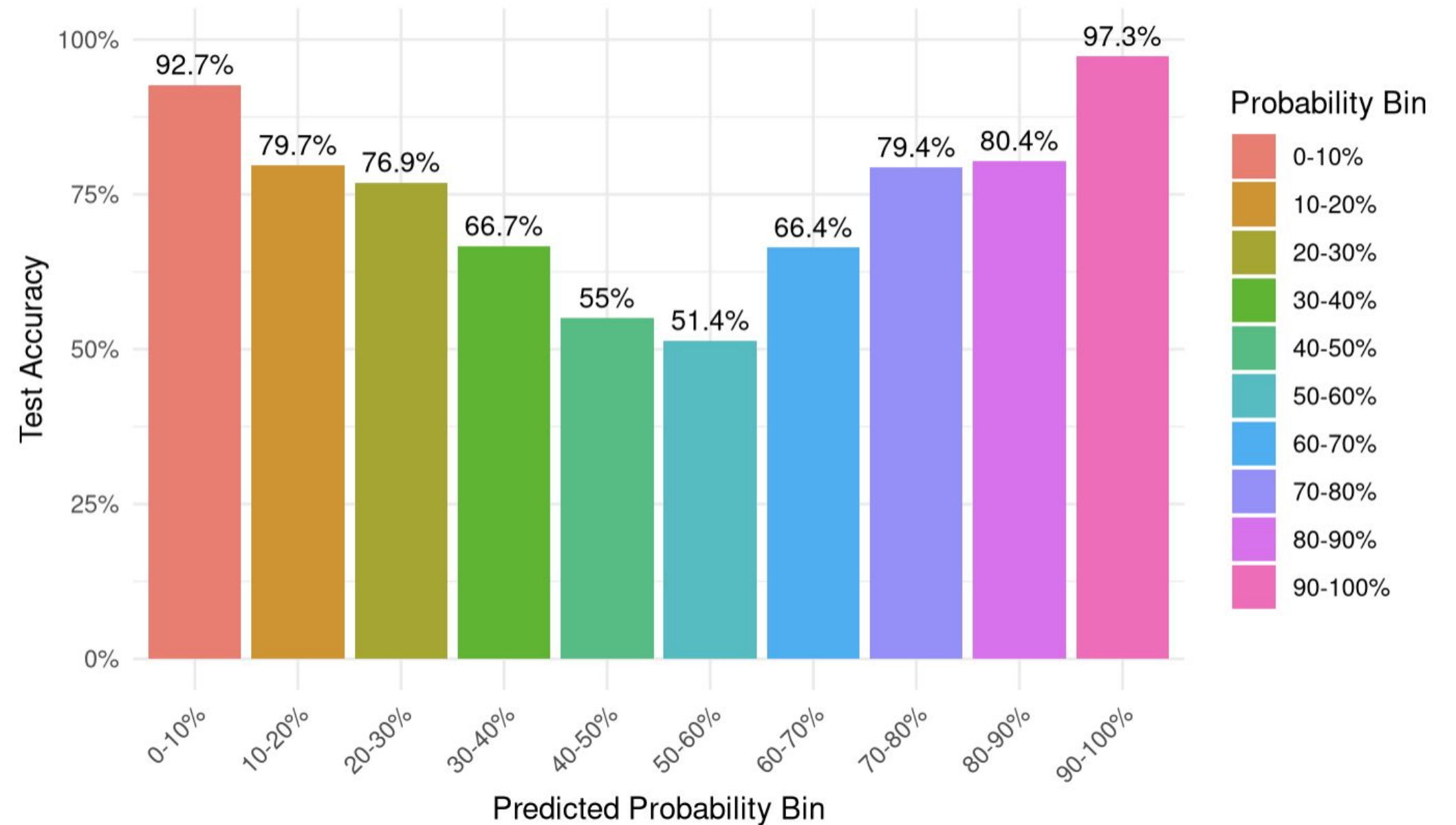
THIRD MODEL

Truncated MOV Reduced Model

Confusion Matrix for Full Non Truncated MOV Model



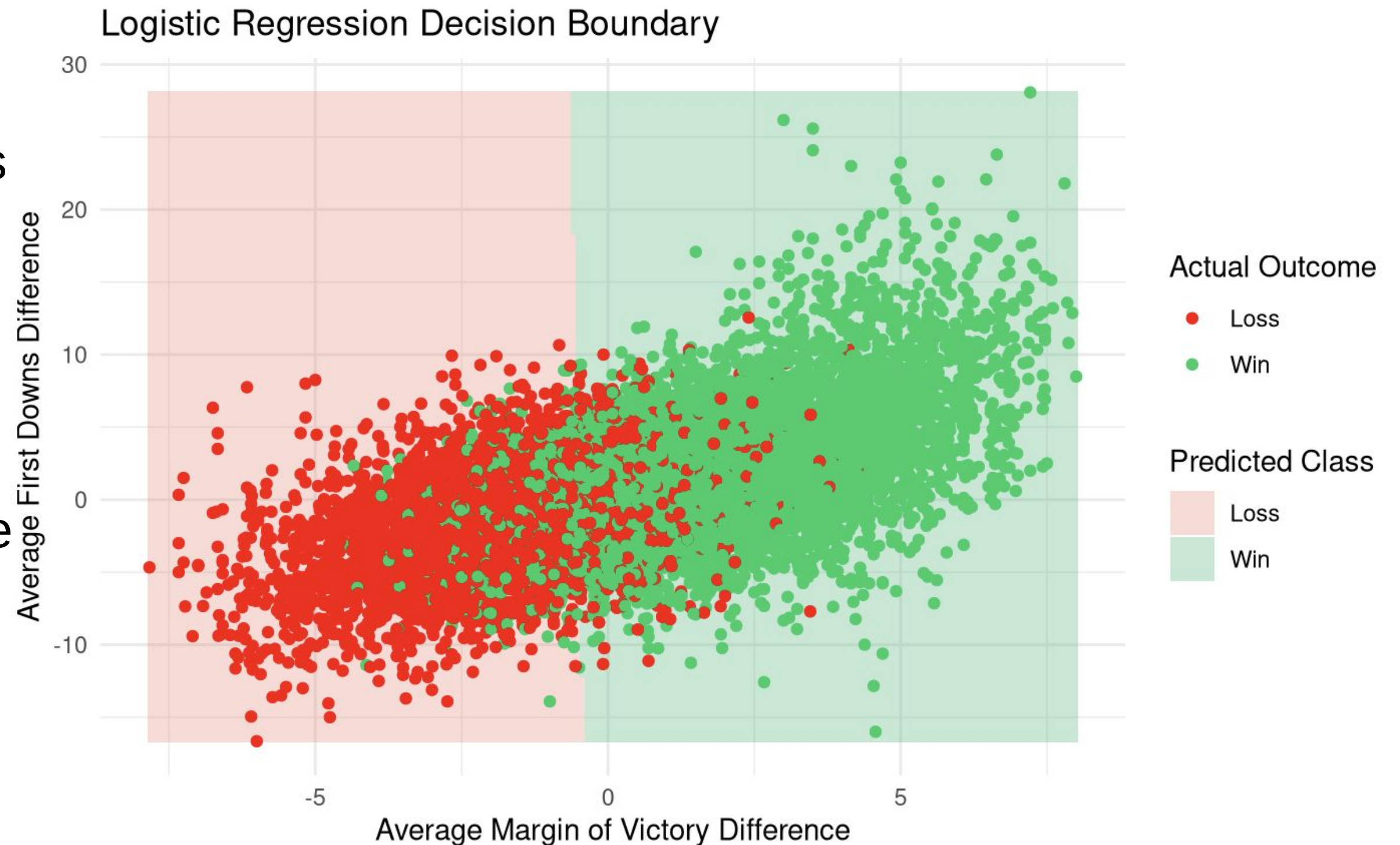
Binned Test Accuracies for Reduced Truncated MOV Model



- Wins Correctly Predicted: 86.98%, Losses Correctly Predicted: 72.97%

THIRD MODEL DECISION BOUNDARY

- Compares the predictions with the two most important variables
- Demonstrates the tendency to overcount the wins

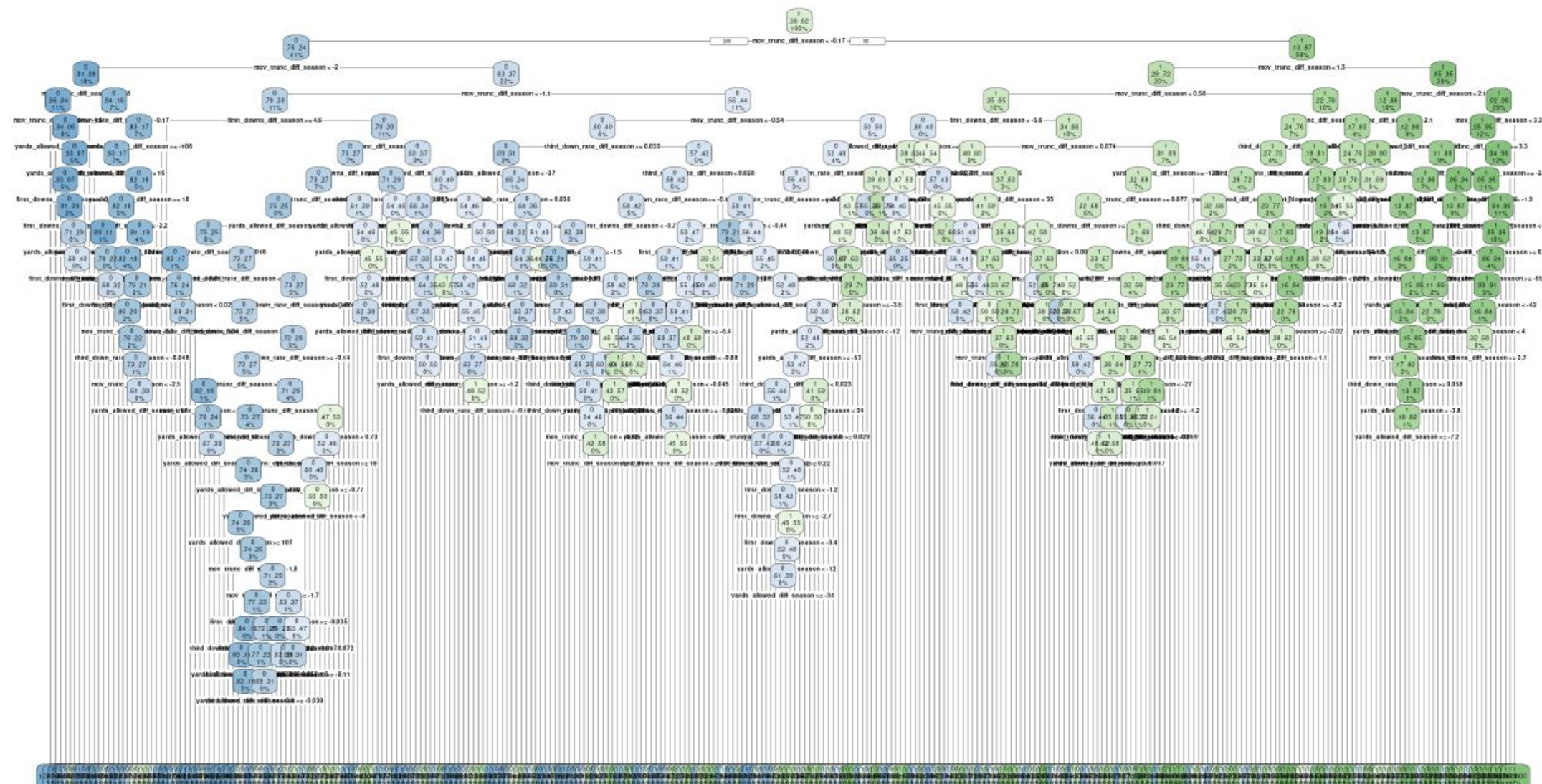


FOURTH MODEL

Decision Tree

- CP value: 0.00001
- Too complex to interpret
- Goal is to prune it down to something more manageable.
- Test Accuracy: 78.51%

Classification Tree College Football Wins

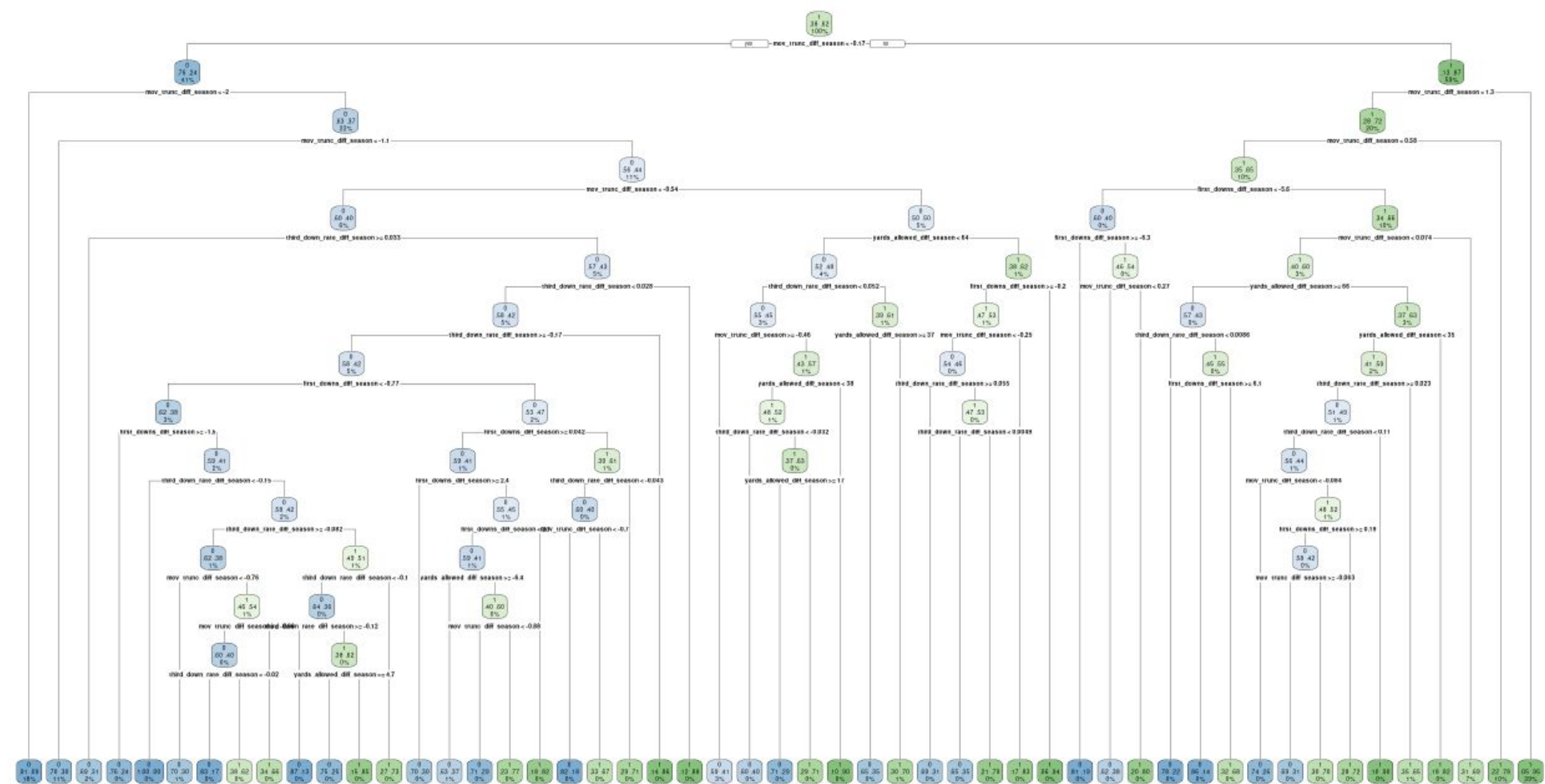


FIFTH MODEL

Pruned Decision Tree

- CP value: 0.00057
- Based on summary output of original tree
- Pruned down but still difficult to interpret
- Test Accuracy: 81.19% (up from 78.51%)

Pruned Classification Tree College Football Wins



Variable importance

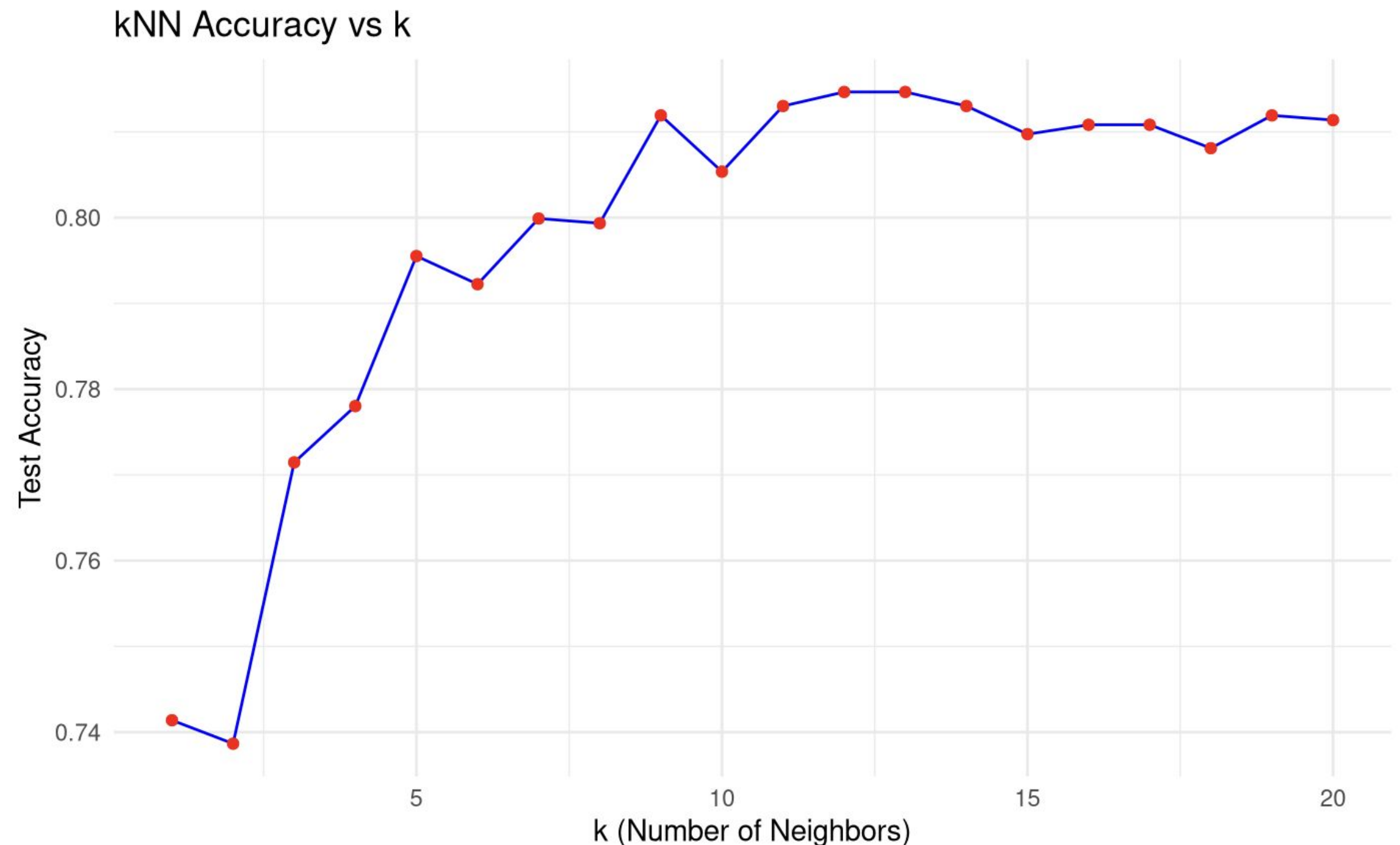
mov_trunc_diff_season
52
first_downs_diff_season
15

yards_allowed_diff_season 17
third_down_rate_diff_season 17

SIXTH MODEL

K-nearest Neighbors

- K value of 13.5 from graph
- Not interpretable
- One of the better test accuracies
- Test Accuracy: 81.52% (up from 81.19%)



MODEL SELECTION

Model	Test Accuracy
Non Truncated Full Log. Regression	78.84%
Truncated Full Log. Regression	81.68%
Truncated Reduced Log. Regression	81.79%
Decision Tree	78.51%
Pruned Decision Tree	81.19%
K Nearest Neighbors	81.52%

PREDICTIONS

Predicting the Six Championship Games

home	away	first_downs_diff_season	third_down_rate_diff_season	mov_trunc_diff_season	yards_allowed_diff_season
JMU	Troy	4.9166666	0.0846663931	2	-126.4
Tulane	North Texas	-3.58333334	-0.0453329644	-0.583333333	-23.5
Texas Tech	BYU	1.91666666	0.09950077461	0.1666666663	-82.8
Alabama	Georgia	-2.333333337	-0.002974443289	-0.75	5.2
Ohio State	Indiana	-1.75	-0.004895104862	0.08333333367	-61.2
Virginia	Duke	-1.083333337	0.09633453791	1.833333333	-109.9

SUN BELT CHAMPIONSHIP

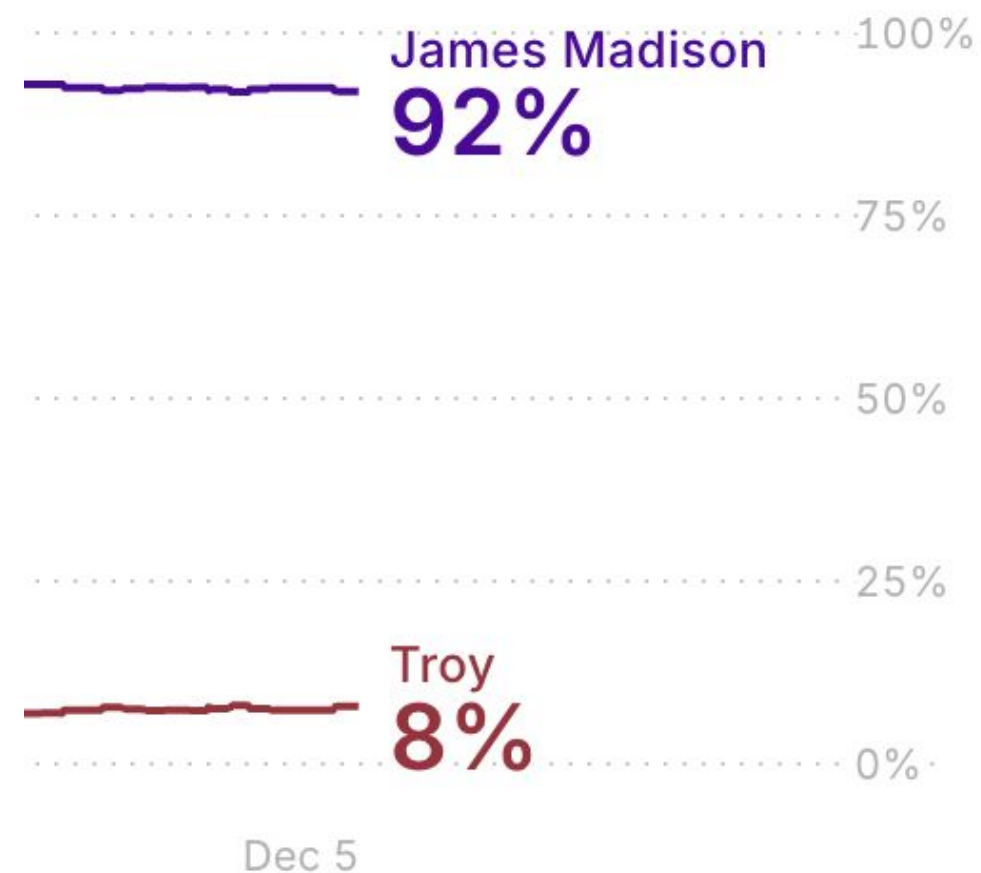
Troy @ JMU

- My model predicts JMU has a 91.05% chance of winning.
- ESPN predicts JMU has a 91.4% chance of winning.
- Kalshi betting markets predicts JMU has a 92% chance of winning

```
##{r}
#JMU Troy Prediction
new_game <- data.frame(
  first_downs_diff_season = 4.916666,
  third_down_rate_diff_season = 0.0846663931,
  mov_trunc_diff_season = 2,
  yards_allowed_diff_season = -126.4
)

pred_prob <- predict(season_glm_MOVTruncRed, newdata = new_game, type = "response")
pred_prob
##
```

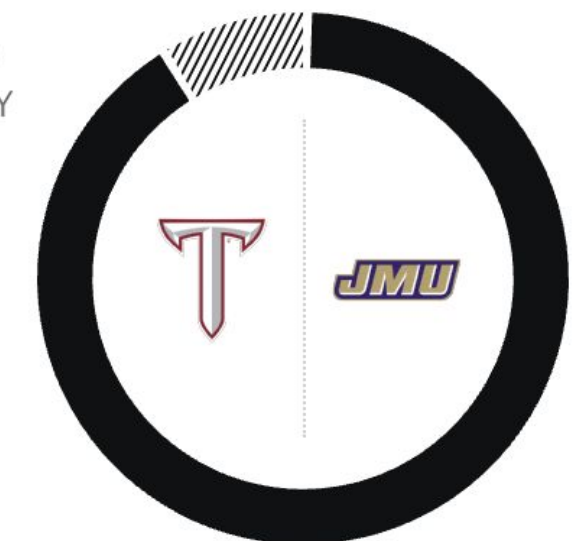
1
0.9104657



MATCHUP PREDICTOR

8.6%
TROY

91.4%
JMU



According to ESPN Analytics



AMERICAN CHAMPIONSHIP

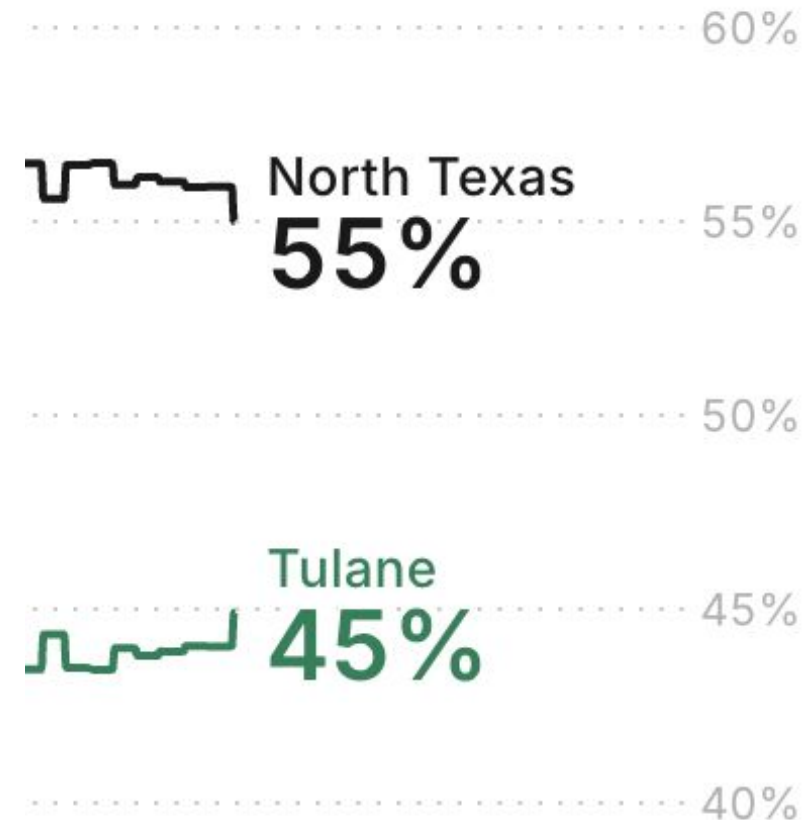
North Texas @ Tulane

- My model predicts Tulane has a 47.63% chance of winning.
- ESPN predicts Tulane has a 42.0% chance of winning.
- Kalshi betting markets predicts Tulane has a 45% chance of winning

```
```{r}
#North Texas Tulane Prediction
new_game <- data.frame(
 first_downs_diff_season = -3.58333334,
 third_down_rate_diff_season = -0.0453329644,
 mov_trunc_diff_season = -0.583333333,
 yards_allowed_diff_season = -23.5
)

pred_prob <- predict(season_glm_MOVTruncRed, newdata = new_game, type = "response")
pred_prob
```
```

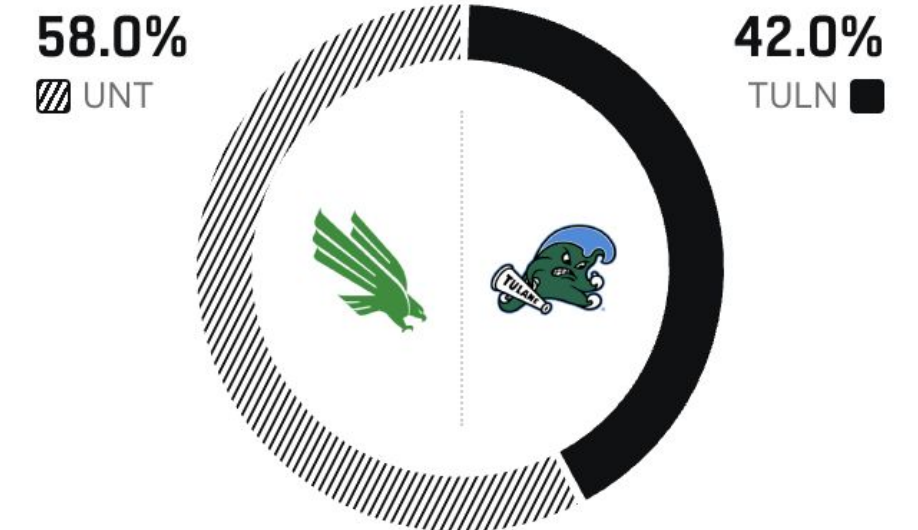
1
0.4762913



Dec 5



MATCHUP PREDICTOR



According to ESPN Analytics



ACC CHAMPIONSHIP

Duke @ Virginia

- My model predicts Virginia has a 89.45% chance of winning.
- ESPN predicts Virginia has a 56.3% chance of winning.
- Kalshi betting market predicts Virginia has a 63% chance of winning

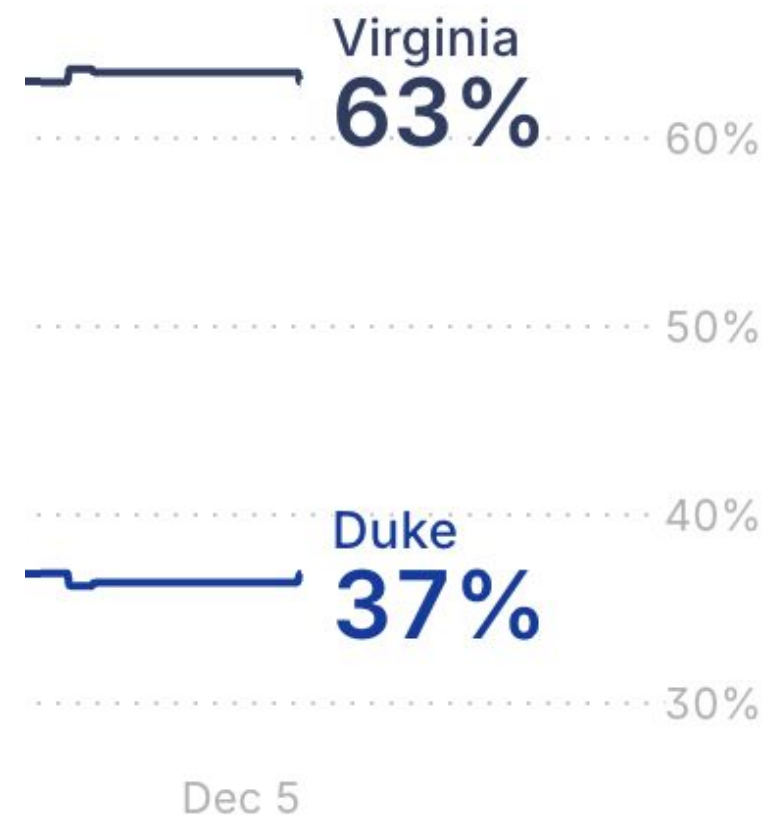
```
```{r}
#Virginia Duke Prediction

new_game <- data.frame(
 first_downs_diff_season = -1.083333337,
 third_down_rate_diff_season = 0.09633453791,
 mov_trunc_diff_season = 1.833333333,
 yards_allowed_diff_season = -109.9
)

pred_prob <- predict(season_glm_MOVTruncRed, newdata = new_game, type = "response")
pred_prob

```
```

1
0.8945296



MATCHUP PREDICTOR



According to ESPN Analytics



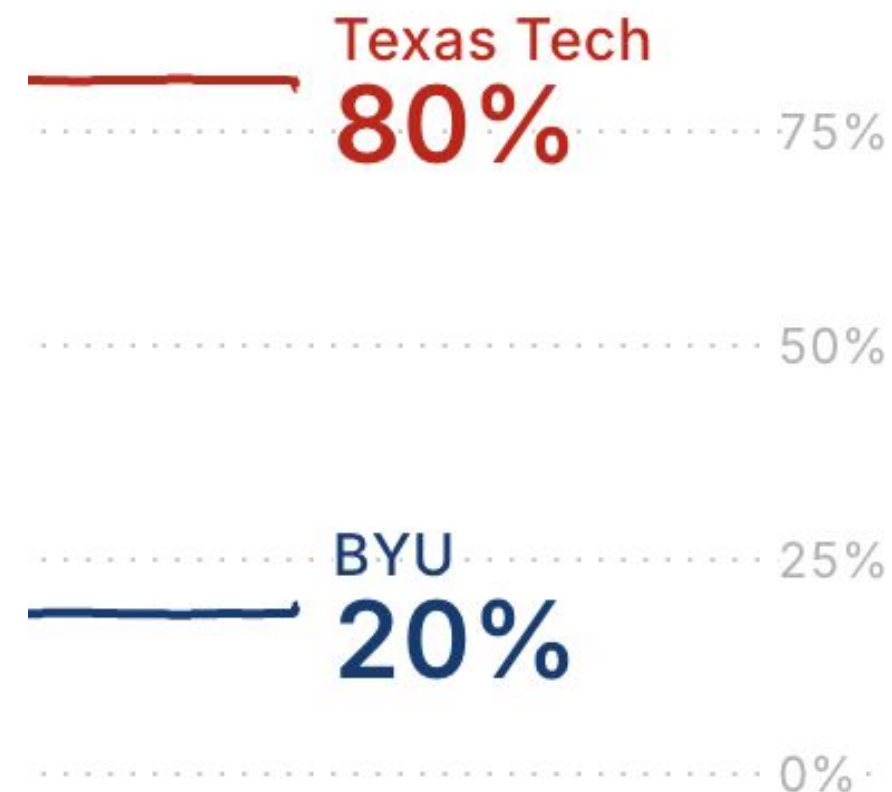
BIG 12 CHAMPIONSHIP

BYU @ Texas Tech

- My model predicts Texas Tech has a 64.71% chance of winning.
- ESPN predicts Texas Tech has a 66.6% chance of winning.
- Kalshi betting market predicts Texas Tech has an 80% chance of winning

```
{r}  
#Texas Tech BYU Prediction  
new_game <- data.frame(  
  first_downs_diff_season = 1.91666666,  
  third_down_rate_diff_season = 0.09950077461,  
  mov_trunc_diff_season = 0.1666666663,  
  yards_allowed_diff_season = -82.8  
)  
  
pred_prob <- predict(season_glm_MOVTruncRed, newdata = new_game, type = "response")  
pred_prob  
...
```

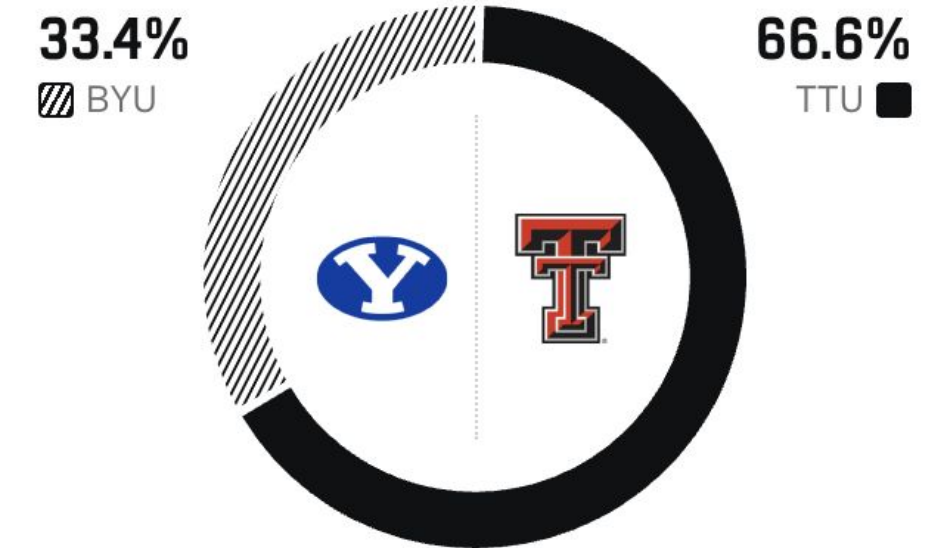
1
0.6470955



Dec 5



MATCHUP PREDICTOR



According to ESPN Analytics



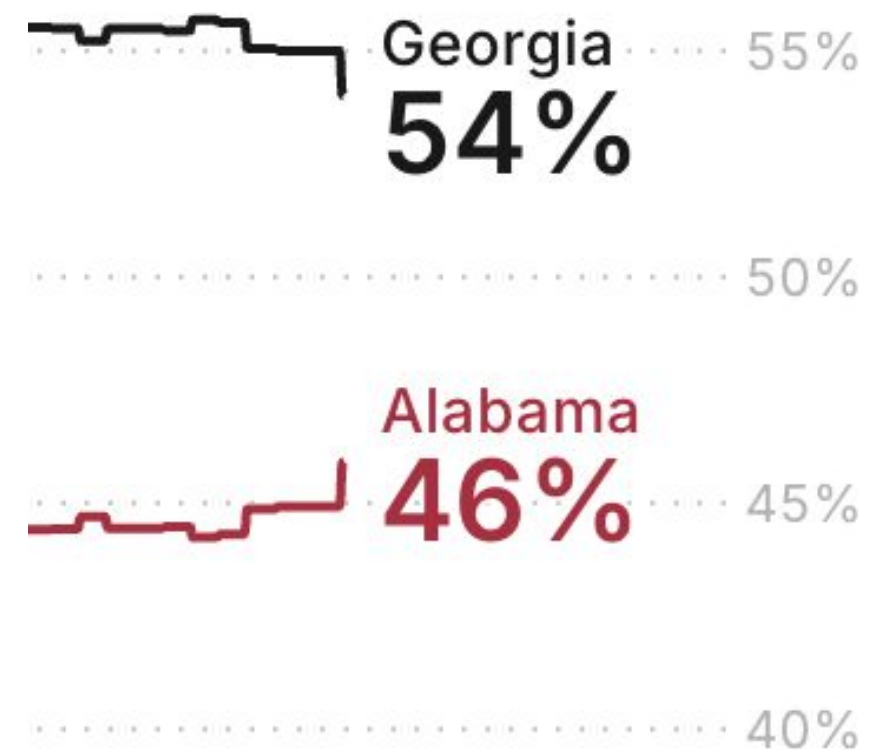
SEC CHAMPIONSHIP

Georgia @ Alabama

- My model predicts Alabama has a 44.01% chance of winning.
- ESPN predicts Alabama has a 51.1% chance of winning.
- Kalshi betting market predicts Alabama has a 46% chance of winning

```
new_game <- data.frame(  
  first_downs_diff_season = -2.333333337,  
  third_down_rate_diff_season = -0.002974443289,  
  mov_trunc_diff_season = -0.75,  
  yards_allowed_diff_season = 5.2  
)  
  
pred_prob <- predict(season_glm_MOVTruncRed, newdata = new_game, type = "response")  
pred_prob  
...
```

1
0.4401363



Dec 5



MATCHUP PREDICTOR



According to ESPN Analytics



BIG TEN CHAMPIONSHIP

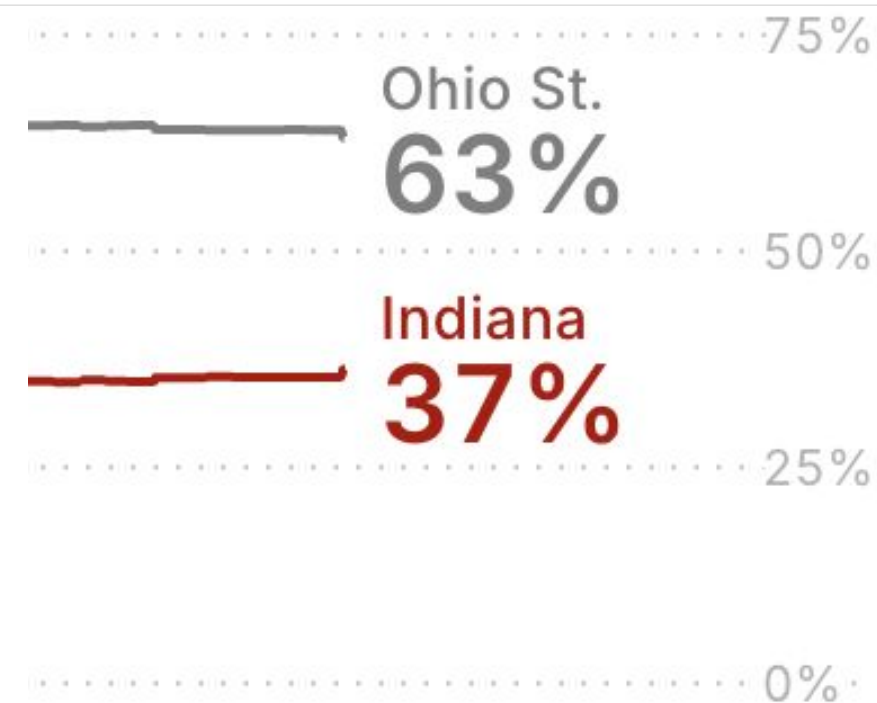
Indiana @ Ohio State

- My model predicts Ohio State has a 62.79% chance of winning.
- ESPN predicts Ohio State has a 49.4% chance of winning.
- Kalshi betting market predicts Ohio State has a 63% chance of winning

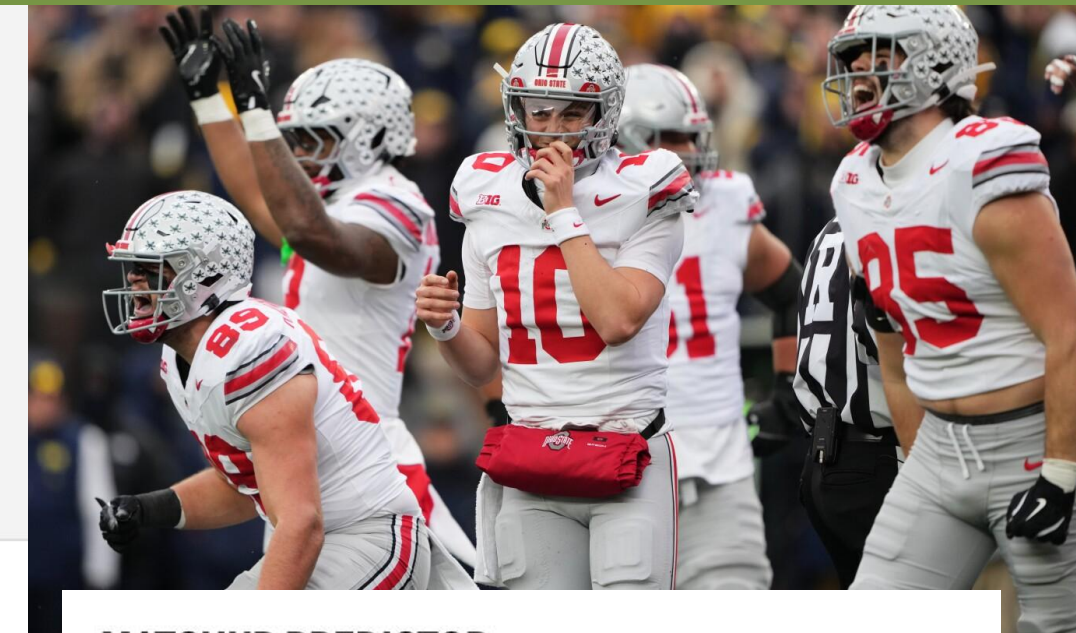
#OHIO ST Indiana Game

```
new_game <- data.frame(  
  first_downs_diff_season = -1.75,  
  third_down_rate_diff_season = -0.004895104862,  
  mov_trunc_diff_season = 0.08333333367,  
  yards_allowed_diff_season = -61.2  
)  
  
pred_prob <- predict(season_glm_MOVTruncRed, newdata = new_game, type = "response")  
pred_prob  
...
```

1
0.6279304



Dec 5



MATCHUP PREDICTOR

50.6%
IU



According to ESPN Analytics



CONCLUSION

Takeaways About the Final Model

- Distribution of test accuracies is what you'd expect from an accurate model.
- Confusion matrix shows the model tends to overpredict wins, leading to a lower prediction accuracy for losses
- Model does much better than in the previous project, going from 71% prediction accuracy to 81.79%.
- Predictions are in line with other methods

